

University of Applied Sciences Hamburg	DEPARTMENT OF ELECTRICAL ENGINEERING	Lab Session 3: Diode
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Lab Session 3: Diode		

Report History

Report 1		Date		Remarks
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Final decision:

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1 Characteristic Curves of Germanium and Silicon Diode

1.1 Theory: Diode Characteristics and Circuit Design

The forward characteristics of a diode show the relationship between the forward current I_F and the forward voltage U_F . When the diode is forward-biased, the current increases exponentially with voltage. This can be modeled by the Shockley diode equation:

$$I_F = I_S \left(e^{\frac{qU_F}{nkT}} - 1 \right)$$

where:

- I_S : reverse saturation current,
- q : electron charge (1.6×10^{-19} C),
- k : Boltzmann constant (1.38×10^{-23} J/K),
- T : temperature in Kelvin,
- n : ideality factor (usually between 1 and 2 for diodes).

To measure this relationship in practice, a simple circuit is used:

- A DC voltage source V_{source} ,
- A resistor R in series with the diode,
- The diode in forward bias.

The forward voltage U_F is measured across the diode, and the current I_F is calculated using:

$$I_F = \frac{V_{\text{source}} - U_F}{R}.$$

1.2 Formulas and Explanation

To solve this task, the following steps and formulas are used:

1. **Calculate the series resistance:** The resistor R limits the current so it stays within the range $0 \leq I_F \leq 10$ mA. Use this formula:

$$R = \frac{V_{\text{source}} - U_F}{I_F}.$$

Here:

- $V_{\text{source}} = 10$ V (maximum DC voltage),
- U_F : forward voltage of the diode,
- I_F : forward current (up to 10 mA).

-
2. **Ensure safe diode operation:** The resistor prevents the current from going too high and damaging the diode. The power dissipated by the diode is:

$$P = U_F \cdot I_F.$$

This value must stay below the diode's maximum power rating.

3. **Find differential resistance:** The differential resistance (AC resistance) of the diode at a given current I_F is calculated using:

$$r_d = \frac{\Delta U_F}{\Delta I_F}.$$

This is the slope of the I_F - U_F curve at a specific point.

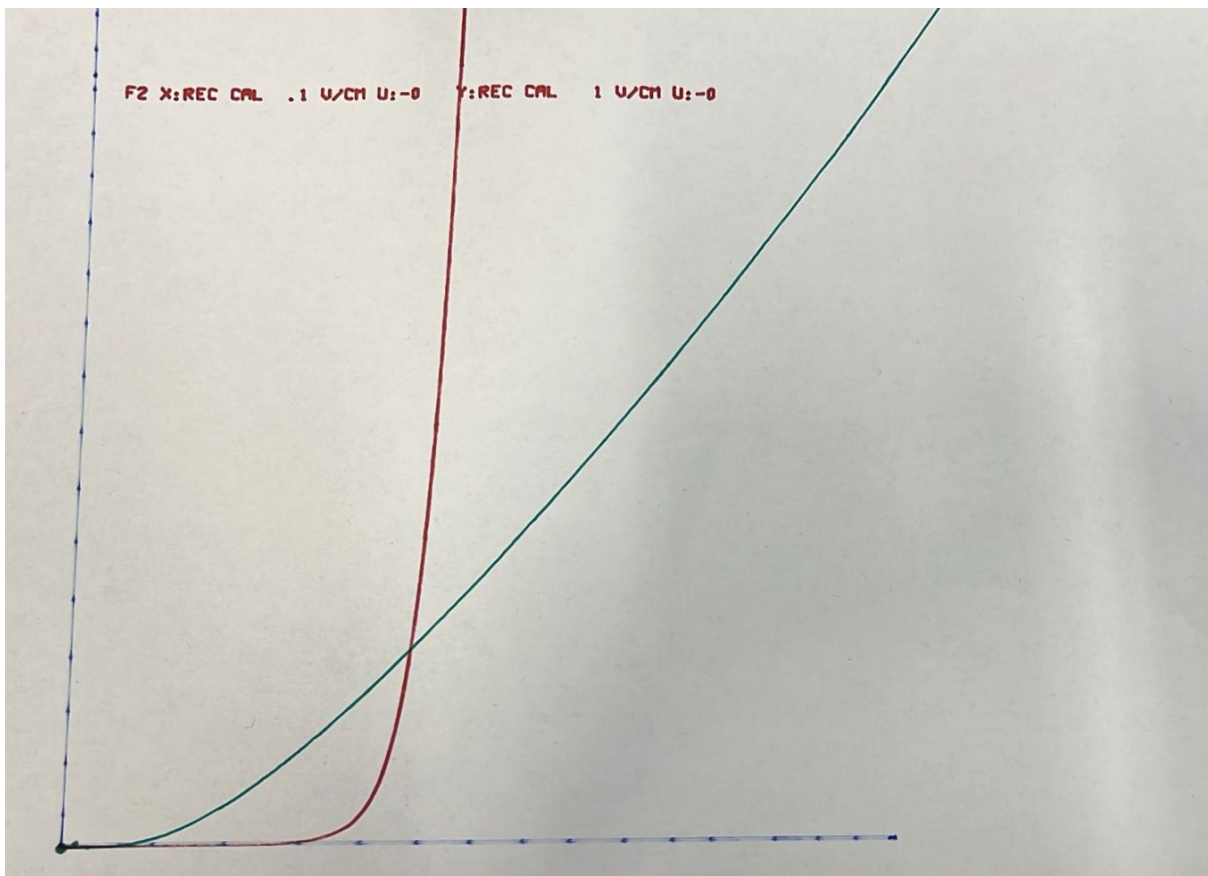


Figure 1: X-Y recorder data

1.3 Procedure

Diode Specifications

1N4148 (Silicon)	
Material	Silicon
Forward Voltage (UF)	Approx. 0.7V at standard forward current
Reverse Breakdown Voltage (VBR)	100V
Maximum Forward Current	300mA (continuous)
Junction Temperature	Up to 175°C
AA138 (Germanium)	
Material	Germanium
Forward Voltage (UF)	Approx. 0.2V to 0.3V at standard forward current
Reverse Breakdown Voltage (VBR)	Approx. 20-30V
Maximum Forward Current	Typically 50mA
Junction Temperature	Lower than silicon diodes (e.g., 75°C)

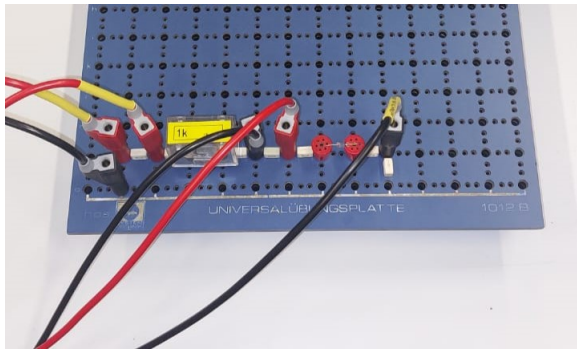


Figure 2: Setup with Germanium Diode

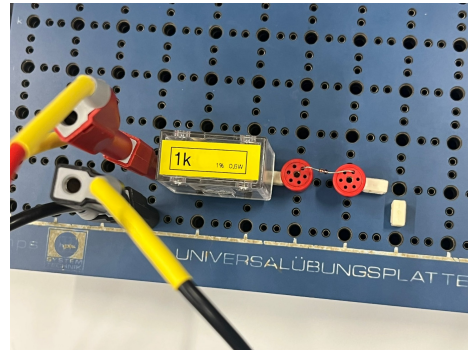


Figure 3: Setup with Silicon Diode

Step 1: Calculate Series Resistance

For $I_F = 0$ to 10 mA, and using approximate forward voltages:

- For the silicon diode ($U_F \approx 0.7$ V):

$$R = \frac{10 \text{ V} - 0.7 \text{ V}}{10 \text{ mA}} = 930 \Omega.$$

- For the germanium diode ($U_F \approx 0.3$ V):

$$R = \frac{10 \text{ V} - 0.3 \text{ V}}{10 \text{ mA}} = 970 \Omega.$$

A standard resistor close to these values, like 1 kΩ, can be used.

Step 2: Safe Diode Operation

The power dissipated by the diode is:

-
- For the silicon diode ($U_F \approx 0.7 \text{ V}$, $I_F = 10 \text{ mA}$):

$$P = 0.7 \text{ V} \cdot 10 \text{ mA} = 7 \text{ mW}.$$

- For the germanium diode ($U_F \approx 0.3 \text{ V}$, $I_F = 10 \text{ mA}$):

$$P = 0.3 \text{ V} \cdot 10 \text{ mA} = 3 \text{ mW}.$$

Step 3: Differential Resistance

To find the differential resistance r_d at $I_F = 2 \text{ mA}$, use the graph of I_F vs. U_F . The resistance is the slope at this point:

$$r_d = \frac{\Delta U_F}{\Delta I_F}.$$

Calculations for diodes:

- The slope for the silicon diode is:

$$r_d = \frac{\Delta U_F}{\Delta I_F} = \frac{0.05 \text{ V}}{0.0015 \text{ A}} = 33.3 \Omega$$

- The slope for the germanium diode is:

$$r_d = \frac{\Delta U_F}{\Delta I_F} = \frac{0.05 \text{ V}}{0.001 \text{ A}} = 50 \Omega$$

2 Characteristic Curve of the Silicon Diode in Semilogarithmic Representation

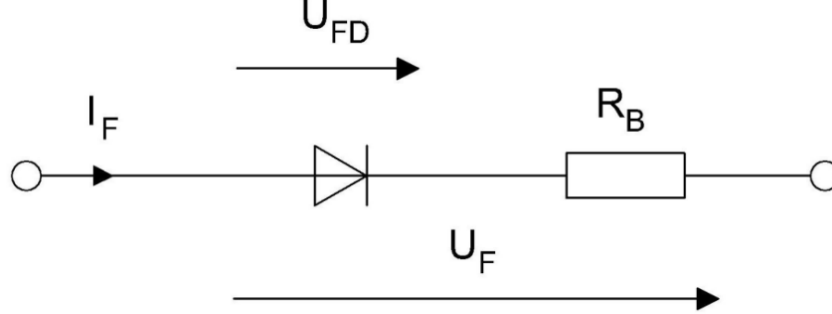


Figure 4: Equivalent Circuit Diagram for a Diode

2.1 Introduction

In this task, we aim to investigate the forward characteristic of the silicon diode 1N4148 by analyzing the relationship between the forward current (I_F) and voltage (U_F). By measuring and plotting these values, we will determine key diode parameters such as reverse saturation current (I_s), bulk resistance (r_b), and the emission coefficient (n). This analysis helps to understand the diode's behavior under forward bias and validates theoretical models with practical data.

2.2 Preparation

2.3 Schottky Equation for Diode Current

The Schottky equation describes the current through a diode as a function of the applied voltage. It is given by:

$$I_F = I_s \left(e^{\frac{U_{FD}}{nV_T}} - 1 \right) \quad (1)$$

Where:

- I_F is the current through the diode.
- I_s is the saturation current.
- U_{FD} is the voltage across the diode.
- n is the ideality factor, typically between 1 and 2.
- $V_T = \frac{kT}{q}$ is the thermal voltage, where k is Boltzmann's constant, T is the absolute temperature, and q is the charge of the electron.

2.4 Current Through the Diode I_F

For a forward-biased diode, the current through the diode is given by:

$$I_F = I_s \left(e^{\frac{U_F D}{n V_T}} - 1 \right) \quad (2)$$

Where u_F is the forward voltage.

2.5 Base Resistor R_b

In a simple diode circuit, a resistor R_b is often connected in series with the diode. The voltage across the resistor is related to the current through the diode and the applied voltage by Ohm's law:

$$U_R = I_F \cdot R_b \quad (3)$$

Where:

- U_R is the voltage across the resistor.
- R_b is the resistance of the base resistor.

The total voltage across the circuit is given by:

$$U_{total} = U_F + U_R = U_F + I_F \cdot R_b \quad (4)$$

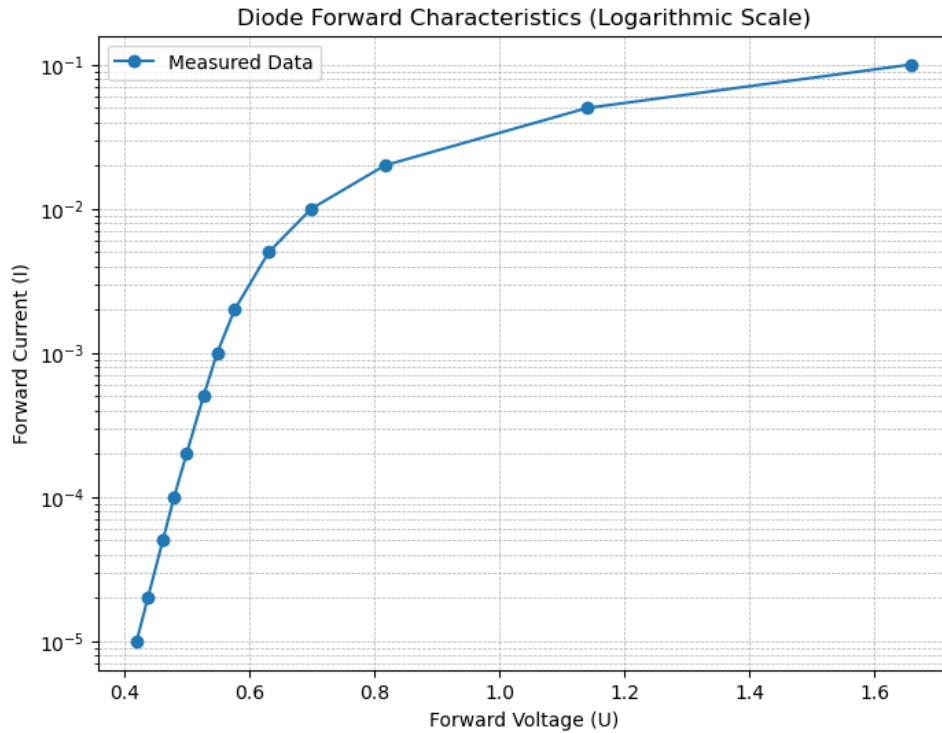
2.6 Procedure

In this experiment, we measured the forward voltage and current across a silicon diode for various applied voltages. The data was collected during the lab session. The results are shown in Table 2.6 below.

Forward Voltage U_{FD} (V)	Forward Current I (I)
0.419 V	10 μ A
0.437 V	20 μ A
0.461 V	50 μ A
0.479 V	100 μ A
0.499 V	200 μ A
0.526 V	500 μ A
0.548 V	1 mA
0.576 V	2 mA
0.6307 V	5 mA
0.6987 V	10 mA
0.8167 V	20 mA
1.1405 V	50 mA
1.6585 V	100 mA

Table 1: Forward Voltage and Current Data

The graph showing the relationship between the forward voltage (V_{FD}) and forward current (I_F) is presented.



2.7 Graphical Analysis of Diode Characteristics

The following graphs show the relationship between the forward voltage (U_F) and the forward current (I_F) for the silicon diode. The first graph represents the measured and theoretical data on a linear scale, while the second one shows the same data on a semilogarithmic scale, highlighting the exponential nature of the diode's behavior.

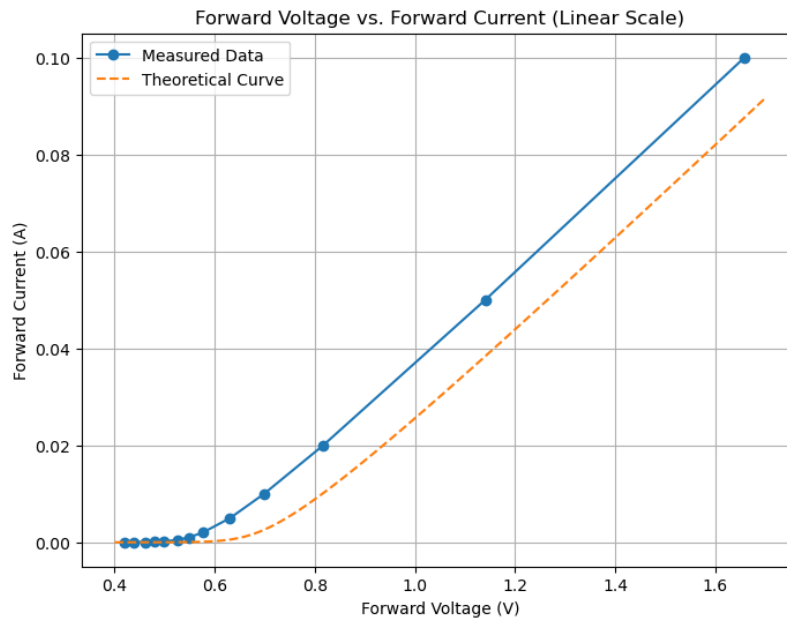


Figure 5: Forward Voltage vs. Forward Current (Linear Scale)

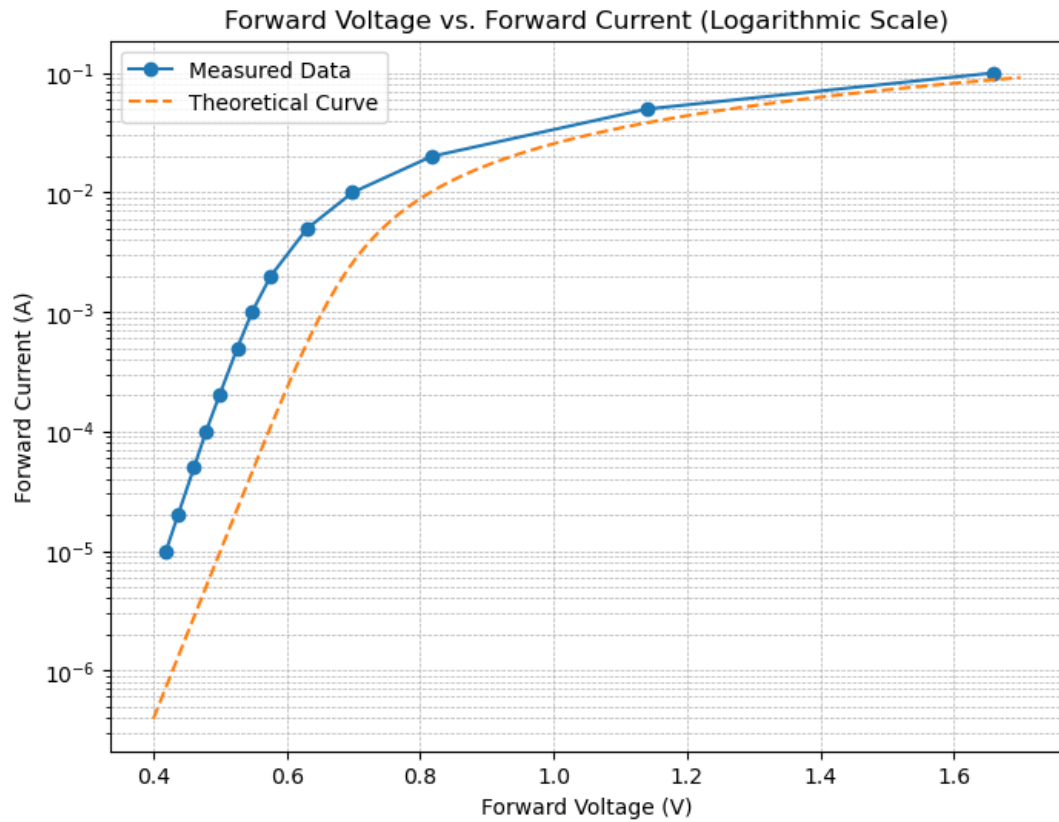


Figure 6: Forward Voltage vs. Forward Current (Logarithmic Scale)

Additionally, the circuit used to measure the diode's current and voltage is shown in the figure below.

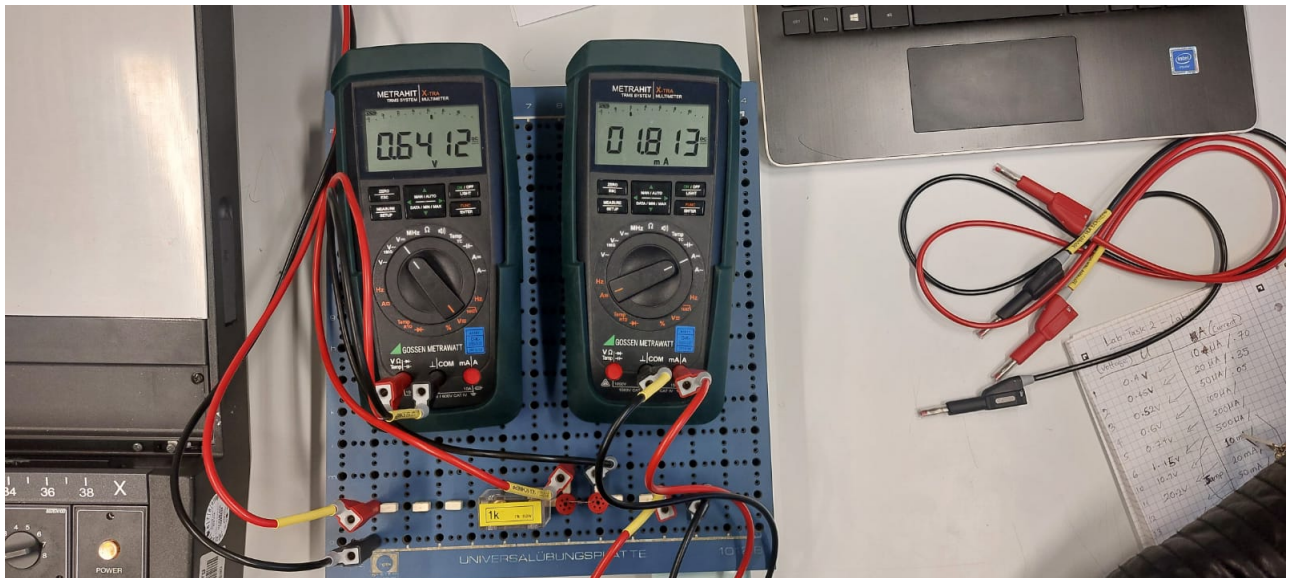


Figure 7: Equivalent Circuit Diagram for a Diode

2.8 Calculation for Differential Resistor r_D

In this section, we will calculate the differential resistance r_D at a forward current $I_F = 2 \text{ mA}$, using the relations and formulas derived in the preparation.

The forward voltage U_F is related to the differential voltage U_{FD} and the base resistor R_B by the equation:

$$U_F = U_{FD} + I_F \cdot R_B$$

Using the Shockley equation, the forward current I_F is given by:

$$I_F = I_s \left(e^{\frac{U_{FD}}{nU_T}} - 1 \right)$$

Where: - I_s is the reverse saturation current, - U_{FD} is the forward voltage across the diode, - n is the ideality factor (emission coefficient), - U_T is the thermal voltage, given by $U_T = 0.025 \text{ V}$ at room temperature.

Step 1: Calculate Reverse Saturation Current I_s The reverse saturation current I_s can be calculated from the Shockley equation by rearranging for I_s :

$$I_s = \frac{I_F}{e^{\frac{U_F}{nU_T}} - 1}$$

For the given data point $I_F = 2 \text{ mA}$ and $U_F = 0.576 \text{ V}$, we can estimate I_s . Assuming an ideality factor $n \approx 1$ and thermal voltage $U_T = 0.025 \text{ V}$, we get:

$$I_s = \frac{2 \times 10^{-3}}{e^{\frac{0.576}{1 \cdot 0.025}} - 1}$$

$$I_s \approx 6.99 \times 10^{-7} \text{ A}$$

Step 2: Calculate the Ideality Factor n To calculate the emission coefficient n , we use the natural logarithmic relationship from the Shockley equation:

$$\ln(I) = \ln(I_s) + \frac{qV}{nkT}$$

Where: - $q = 1.602 \times 10^{-19} \text{ C}$ is the charge of an electron, - $k = 1.38 \times 10^{-23} \text{ J/K}$ is Boltzmann's constant, - $T = 300 \text{ K}$ is the absolute temperature.

By plotting $\ln(I)$ vs. U for the available data and determining the slope, the emission coefficient n can be determined. From the graph, the slope was found to be approximately 38.5.

Using the slope, we can rearrange the formula to solve for n :

$$n = \frac{q}{\text{slope} \cdot kT}$$

Substituting the values:

$$n = \frac{1.602 \times 10^{-19}}{38.5 \times (1.38 \times 10^{-23}) \times 300}$$

$$n \approx 1.005$$

This result aligns closely with the theoretical value of $n \approx 1$ for a silicon diode.

Step 3: Calculate Forward Voltage U_{FD} Now that we have the value of I_s and n , we can calculate the forward voltage U_{FD} at $I_F = 2 \text{ mA}$ using the rearranged Shockley equation:

$$U_{FD} = n \cdot U_T \cdot \ln \left(\frac{I_F}{I_s} + 1 \right)$$

Substituting the values:

$$U_{FD} = 1 \cdot 0.025 \cdot \ln \left(\frac{2 \times 10^{-3}}{6.99 \times 10^{-7}} + 1 \right)$$

$$U_{FD} = 0.025 \cdot \ln(2000.699)$$

$$U_{FD} \approx 0.025 \cdot 7.955936 = 0.198984 \text{ V}$$

Step 4: Calculate Base Resistor R_B The bulk resistance R_B can be calculated using the formula:

$$R_B = \frac{U_F - U_{FD}}{I_F}$$

Substituting the values $U_F = 0.9735 \text{ V}$, $U_{FD} = 0.198984 \text{ V}$, and $I_F = 2 \text{ mA} = 0.002 \text{ A}$:

$$R_B = \frac{0.9735 - 0.198984}{0.002}$$

$$R_B = \frac{0.774516}{0.002} = 7.903 \Omega$$

Thus, the calculated bulk resistance is:

$$R_B = 7.903 \Omega$$

3 Half-wave rectifier circuit

3.1 Introduction

The ripple factor is a measure of the effectiveness of rectification in reducing the AC components in a rectified DC output. It is defined as the ratio of the RMS value of the AC component to the DC value of the output voltage. In this experiment, the ripple factor for a half-wave rectifier circuit is analyzed using two different capacitance values: $C_L = 47\text{ nF}$ and $C_L = 470\text{ nF}$.

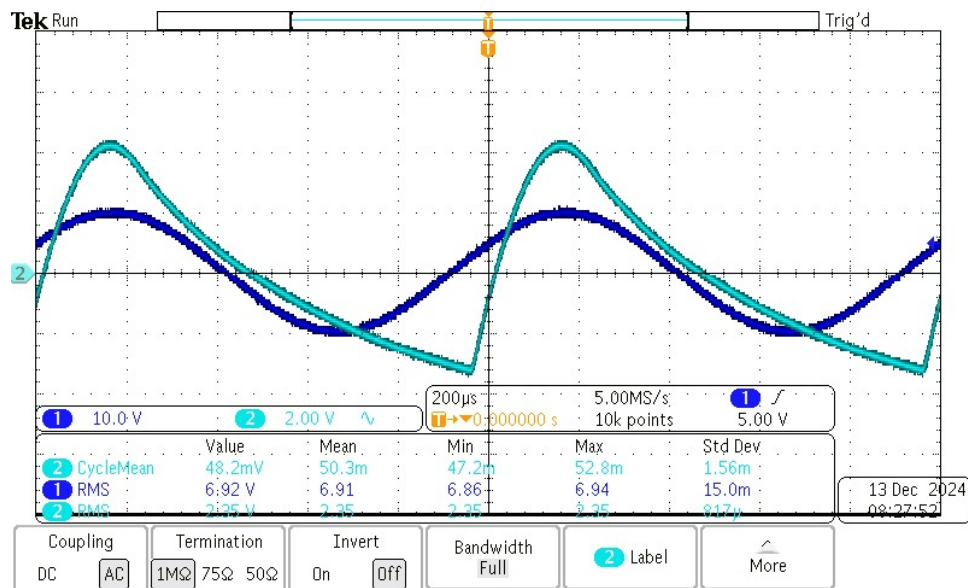


Figure 8: U1 and U2 for the 47nF Capacitor

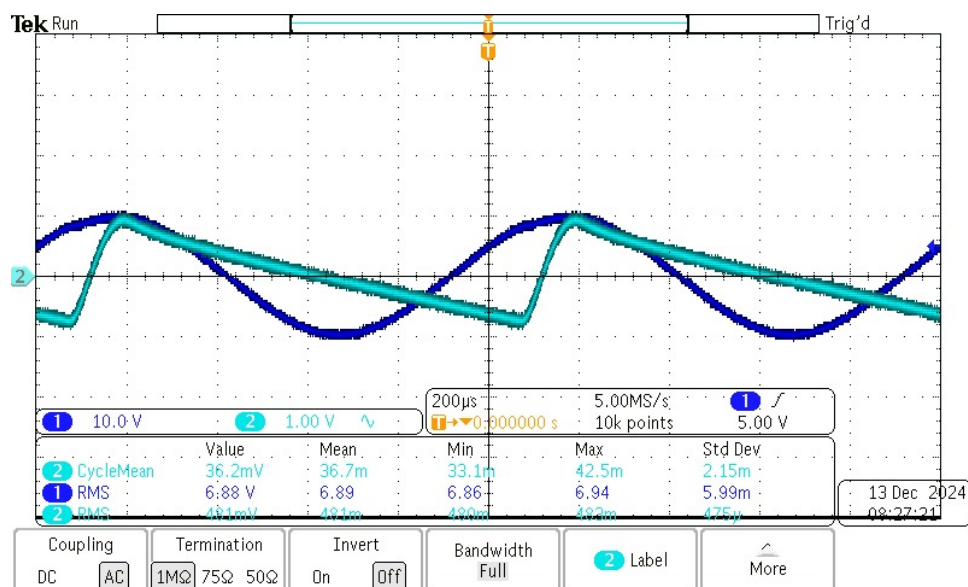


Figure 9: U1 and U2 for the 470nF Capacitor

3.2 Ripple Factor Calculation

The ripple factor (γ) can also be calculated using the formula:

$$\gamma = \sqrt{\left(\frac{V_{\text{rms}}^2}{V_{\text{dc}}^2} - 1\right)}$$

3.2.1 Given Values

For the given experiment, the following values are provided:

- For $C = 470 \text{ nF}$:
 - $V_{\text{rms}} = 6.88 \text{ V}$
 - $V_{\text{dc}} = 36.2 \text{ mV} = 0.0362 \text{ V}$
- For $C = 47 \text{ nF}$:
 - $V_{\text{rms}} = 6.92 \text{ V}$
 - $V_{\text{dc}} = 48.2 \text{ mV} = 0.0482 \text{ V}$

3.3 Ripple Factor Calculation

Now, we calculate the ripple factor for each case using the formula:

$$\gamma = \sqrt{\left(\frac{V_{\text{rms}}^2}{V_{\text{dc}}^2} - 1\right)}$$

3.3.1 For $C = 470 \text{ nF}$:

Substitute the given values:

$$\gamma = \sqrt{\left(\frac{6.88^2}{0.0362^2} - 1\right)} \approx \sqrt{\left(\frac{47.3344}{0.00131144} - 1\right)} \approx \sqrt{36,073.7 - 1} \approx \sqrt{36,072.7} \approx 190.1$$

3.3.2 For $C = 47 \text{ nF}$:

Substitute the given values:

$$\gamma = \sqrt{\left(\frac{6.92^2}{0.0482^2} - 1\right)} \approx \sqrt{\left(\frac{47.8864}{0.00232324} - 1\right)} \approx \sqrt{20,612.1 - 1} \approx \sqrt{20,611.1} \approx 143.0$$